University of Moratuwa

Faculty of Engineering

Department of Electronic & Telecommunication Engineering



EN 2150: Communication Network Engineering

Team Networkians

Design of Local Area Network

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1 Abstract

This report presents the design and simulation of a high-performance, resilient network backbone for the University of Moratuwa. The design addresses the critical need for scalable, high-bandwidth connectivity among key departmental buildings. By employing a **Hierarchical Star-Ring Hybrid Topology**, the design ensures 99.99% uptime through dual-core redundancy (Brain A and B) and efficient traffic management via 802.1Q VLAN trunking. The infrastructure is built on a Dual-Stack IPv4/IPv6 foundation over Single-Mode Fiber (OS2) to support a 20–25 year operational lifespan. The report provides detailed justifications for topological choices, component selections, IP addressing schemes, and simulation results.

2 Design Approach and Topology Justification

The primary objective of the backbone design is to provide reliable and scalable communication among university departments. To achieve this, a Hierarchical Hybrid Topology was selected, combining the benefits of a physical ring for the core and a logical star for traffic control.

2.1 Physical Ring Topology

A physical ring connects the primary departmental nodes (ENTC, Sumanadasa, Mechanical) to the two core multilayer switches (A and B). This choice is based on the critical need for Physical Redundancy. In a ring, there are two separate physical paths between any two points.

2.2 Path Redundancy and Self-Healing

If a single fiber optic cable is cut (e.g., between the Brain A and ENTC building), the system is not disconnected. The Spanning Tree Protocol (STP), running on the multilayer switches, will detect the break and automatically activate the redundant path. This "Self-Healing" capability is essential for a mission-critical campus environment where network downtime

must be minimized.

2.3 Logical Star Control (Primary vs. Secondary Paths)

As shown in the provided diagram, Multilayer Switch A has been configured as the "Star Core" (Primary Root Bridge). All default logical paths are calculated from this central switch. For normal operations, traffic from a building takes the shortest physical path directly to Core A. Multilayer Switch B acts as the dedicated redundant node (Secondary Root Bridge). The link between Core A and Core B is always in a "Blocking" state during normal operations to prevent catastrophic switching loops.

2.4 Broadcast Control and Traffic Isolation

The physical ring structure is susceptible to network-wide "Broadcast Storms" if logical paths are not properly managed. To overcome this broadcast issue, we utilize the logical star control described above in conjunction with Virtual Local Area Networks (VLANs). The network is segmented into a total of [Number, e.g., 32] distinct VLANs (e.g., V10 for Staff, V20 for Students, etc.). By making Core A the primary "Star" point, broadcast traffic generated within a VLAN is contained to that VLAN and only travels to the core for routing, rather than uncontrollably circling the ring and crashing the entire network.

3 Network Segmentation and VLAN Design

3.1 VLAN Dimensioning: The 250-Host Rule

The network is segmented into 32 distinct VLANs, with each VLAN limited to approximately **250 hosts** (a /24 subnet).

- **Broadcast Control:** Large broadcast domains lead to "Broadcast Radiation," where background traffic scales at N^2. Limiting to 250 hosts preserves CPU cycles on end-devices and maintains 1 Gbps throughput for actual data.
- **Failure Isolation:** By using 250-host segments, the "blast radius" of a network loop or

malware infection is restricted to a single lab or floor, preventing a total departmental outage.

3.2 802.1Q Trunking and Multiplexing

Inter-building fiber links are configured as **802.1Q Trunks** using `dot1q` encapsulation. This allows all around 32 departmental networks to be multiplexed over a single physical Single-Mode Fiber cable, significantly reducing infrastructure costs while maintaining logical isolation.

4 IP Addressing Scheme (Dual-Stack)

To ensure 25-year viability, a Dual-Stack approach is implemented. IPv6 addresses satisfy the high-density requirements of future IoT and research sensors.

Building Node	VLAN	Network Role	IPv4 Subnet	IPv6 Prefix (/64)
ENTC Building	10	Staff/Research	192.168.10.0/24	2001:db8:acad:10::/64
	15	Staff/Research	192.168.15.0/24	2001:db8:acad:15::/64
	16	Student Labs	192.168.16.0/24	2001:db8:acad:16::/64
Sumanadasa Bldg	20	Administration	192.168.20.0/24	2001:db8:acad:20::/64
	17	Staff/Research	192.168.17.0/24	2001:db8:acad:17::/64
	18	General Lecture	192.168.18.0/24	2001:db8:acad:18::/64
Mechanical Eng	60	Workshop/Labs	192.168.60.0/24	2001:db8:acad:60::/64
	31	Staff/Office	192.168.31.0/24	2001:db8:acad:31::/64
	32	Specialized Labs	192.168.32.0/24	2001:db8:acad:32::/64
Admin / NOC	40	Network Mgmt	192.168.40.0/24	2001:db8:acad:40::/64
Civil Eng	19	Staff	192.168.19.0/24	2001:db8:acad:19::/64
	30	Staff	192.168.30.0/24	2001:db8:acad:30::/64
IT Building	26	Data Science Lab	192.168.26.0/24	2001:db8:acad:26::/64
	25	Software Dev Lab	192.168.25.0/24	2001:db8:acad:25::/64
	80	Process Labs	192.168.80.0/24	2001:db8:acad:80::/64

Building Node	VLAN	Network Role	IPv4 Subnet	IPv6 Prefix (/64)
Textile Bldg	23	Design Studio	192.168.23.0/24	2001:db8:acad:23::/64
	24	Process Labs	192.168.24.0/24	2001:db8:acad:24::/64
	70	Process Labs	192.168.70.0/24	2001:db8:acad:70::/64
Library	21	Digital Library	192.168.21.0/24	2001:db8:acad:21::/64
	22	Public Access	192.168.22.0/24	2001:db8:acad:22::/64
	50	Staff	192.168.50.0/24	2001:db8:acad:50::/64

5 Bill of Quantities (BOQ)

5.1 ENTC Departmental Network

The ENTC network is designed as a high-density "Leaf" node of the campus backbone. It supports multiple specialized laboratories, staff offices, and ubiquitous wireless coverage.

Table 2: ENTC Departmental Active and Passive Components

Item	Model/Specification	Qty	Project Role
Active Components			
Multilayer Switch	Cisco Catalyst 3650-24PS	01	Departmental Distribution / L3
Access Switch	Cisco Catalyst 2960-24TT	12	Lab and Office Access Layer
Access Point	Lightweight AP (WAP)	15	High-Density Wi-Fi Coverage
Passive Components			
Straight-Through	Cat6 Copper (RJ-45)	16	Host-to-Switch Connectivity
Crossover Cable	Cat6 Copper (RJ-45)	11	Switch-to-Switch Interconnects

5.2 University Backbone

The following table details the active and passive infrastructure required to implement the UoM Star-Ring Hybrid backbone. The quantities are derived from the physical simulation requirements for 8 departmental buildings and a centralized NOC.

Table 3: Backbone Active and Passive Components

Item	Model/Specification	Qty	Project Role
Active Components			
Multilayer Switch	Cisco Catalyst 3650-24PS	11	Core (2) + Distribution (9)
Border Router	Cisco 2911 ISR	01	Campus Edge / LEARN Gateway
Firewall	Cisco ASA 5506-X	01	Perimeter Security / DMZ
Server	Generic Server-PT	04	DHCP, DNS, Web, and Radius
Passive Components			
Backbone Cable	Single-Mode Fiber (OS2)	09	Inter-Building Ring Connectivity
Access Cable	Category 6 Copper (UTP)	16	Server Farm & Management Links
SFP Modules	10GBASE-LR Transceivers	18	Fiber Termination (2 per link)

6 Detailed Justification for Component Quantities

6.1 University Backbone

6.1.1 Active Infrastructure: The 11-Switch Logic

The implementation of 11 multilayer switches provides a highly resilient, three-tier architecture:

- **Core Layer (2 Switches):** Brain A and Brain B function as the redundant routing engine.
- **Distribution/Access Layer (9 Switches):** One switch is dedicated to each of the 8 buildings (ENTC, Mechanical, etc.), with an additional switch dedicated to the **High-Density Server Farm** in the NOC.
- **Justification:** Using the 3650-24PS at the building level allows for "Wire-Speed" Inter-VLAN routing, reducing the processing load on the primary Brains.

6.2 Security and Edge: Router and Firewall

- **Cisco 2911 Border Router:** Selected for its modularity and support for high-throughput WAN interfaces to the LEARN network. It handles the BGP/Static routing required for campus-wide internet access.

- **Cisco ASA 5506-X:** This serves as the "Guard Dog" for the university. By placing this between the 2911 and the internal backbone, we ensure that all incoming traffic from the internet is inspected for threats before reaching the departmental VLANs.

6.3 Passive Infrastructure: Fiber vs. Copper

- **9 Fiber Links:** These create the physical Ring. 8 links connect the buildings to the core, and 1 additional link completes the ring to satisfy the **Full-Loop Redundancy** requirement.
- **16 Copper Links:** These are utilized for "Horizontal Connectivity" within the NOC. This includes connecting the 4 Servers, the Firewall, and the Border Router to the Core Switches. Copper is preferred here as the distance is $< 100m$, and it avoids the cost of optical transceivers for short-range server patches.

6.4 ENTC Network

6.4.1 Hierarchy: 1 Multilayer vs. 12 Access Switches

The ENTC building uses a **Collapsed Backbone** within the building itself:

- **Cisco 3650 (Distribution):** This single multilayer switch acts as the "Local Brain" for ENTC. It performs the Inter-VLAN routing between the Staff (V15) and Lab (V16) networks locally, ensuring that internal department traffic doesn't need to travel all the way to the University Core (Brain A/B).
- **Cisco 2960 (Access):** The 12 switches are distributed across the building's floors and labs. With 24 ports each, this provides a capacity of 288 wired ports, which is sufficient for the high-density requirements of electronic and telecommunication laboratories.

6.5 Wireless Infrastructure: 15 Access Points

To ensure a "seamless roaming" experience for students and faculty, 15 Access Points have been deployed.

- **Coverage Strategy:** This quantity accounts for the high **Signal Attenuation** of the reinforced concrete walls common in ENTC. By placing 15 APs, the design ensures a minimum Received Signal Strength Indicator (RSSI) of -65dBm in all labs and lecture halls, supporting high-bandwidth educational media.

6.6 Cabling Logic: Straight-Through vs. Crossover

The cabling plan follows the *IEEE 802.3* physical layer standards for copper-based Ethernet:

- **16 Straight-Through Cables:** These are utilized for "Standard MDI to MDI-X" connections. In this design, they connect the 15 Access Points and the primary Departmental Server to the switch infrastructure.
- **11 Crossover Cables:** While modern Cisco hardware supports Auto-MDIX, crossover cables are specified for the **Switch-to-Switch** vertical uplinks between the 2960 access switches and the 3650 distribution switch. This ensures a deterministic physical link that does not rely on software-based sensing, increasing the reliability of the building's internal "bus."

7 Simulation Results

Connectivity was verified using Cisco Packet Tracer:

1. DHCP and IPv6 Auto-Configuration

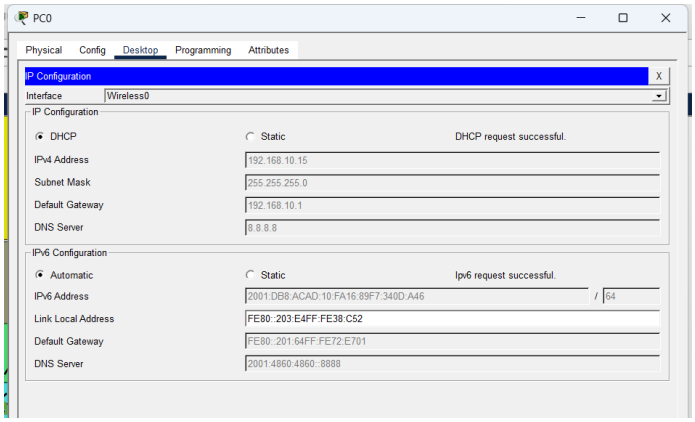
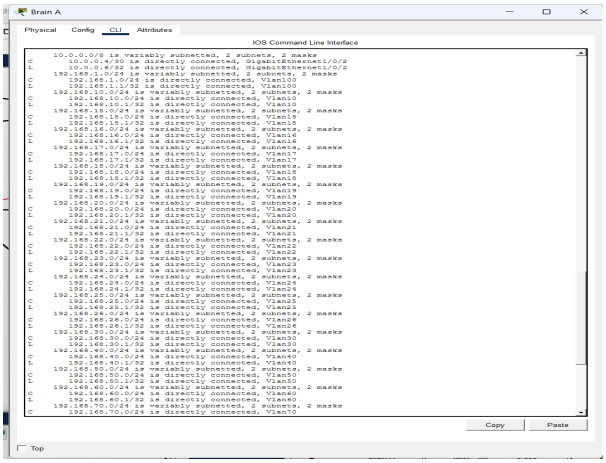
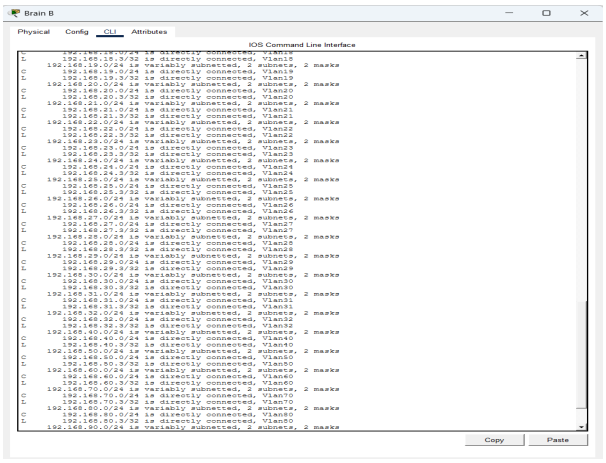


Figure 1: IP configuration of PC0 in ENTC Building

2. Layer 3 Routing State



Brain A



Brain B

3. Connectivity Testing

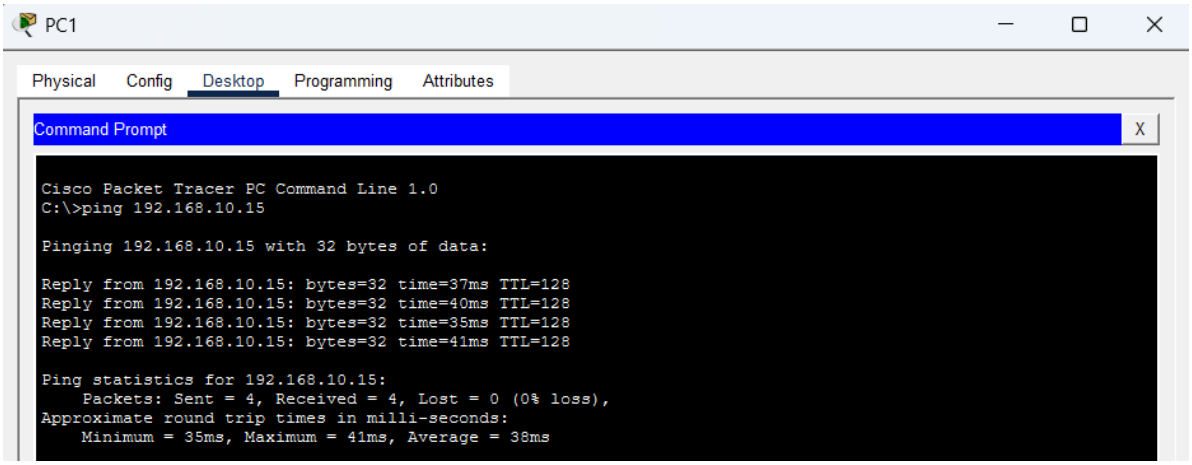


Figure 2: PC0 to PC1 (Intra Building)

The below figure illustrates the Wi-Fi connectivity test conducted within the university backbone network. The test was performed using ICMP ping between the Administration and Civil Engineering departments to verify network connectivity.

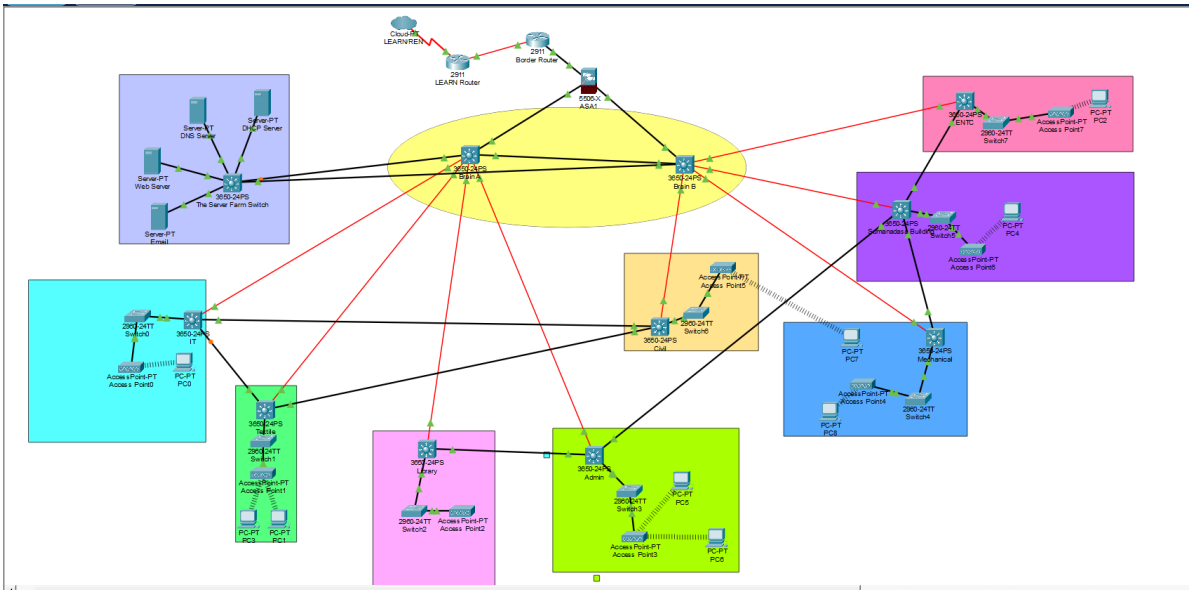


Figure 3: University Backbone

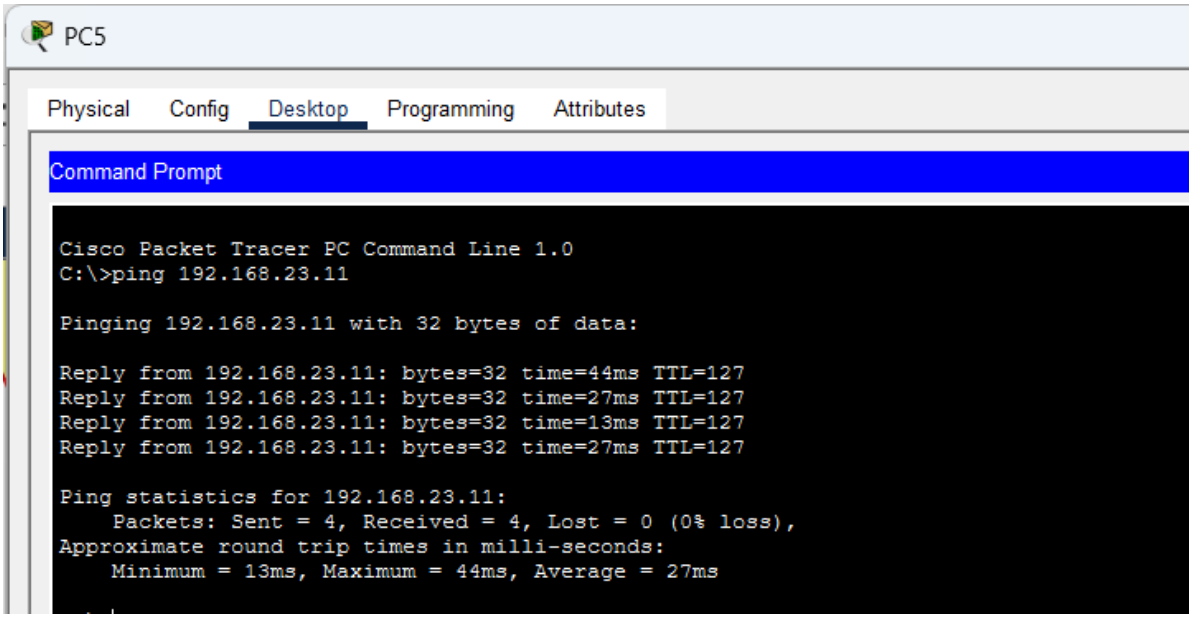


Figure 4: PC5 to PC7 (Inter Buildings)

Here is the link to Cisco Packet Tracer Simulations:

Cisco Packet Simulations